

Effect Sizes & Statistical Measures

Method Dossier | Basic Level

1. Method Description

Effect sizes quantify practical significance, independent of sample size. Cohen conventions: $d=0.2$ small, 0.5 medium, 0.8 large. $r=0.1$ small, 0.3 medium, 0.5 large. η^2 : 0.01 small, 0.06 medium, 0.14 large.

d-family (mean differences): Cohen d , Hedges g , d_z (paired)

r-family (correlations): Pearson r , η^2 , partial η^2 , Cramers V

Risk: OR, RR, ARR, NNT

2. Examples

Medicine: ARR=0.10, NNT=10, RR=0.5, OR=0.44

Psychology: $d=0.75$, 95% CI [0.42,1.08], Hedges $g=0.74$

3. Code Examples

R:

```
cohens_d(dv~gruppe, data=df)
etaSquared(aov_mod)
1/ARR # NNT
```

Python:

```
pg.compute_effsize(x, y, eftype="cohen")
```

4. Exercises

1. $n_1=25$, $n_2=25$, $M_1=85$, $M_2=72$, $SD_{\text{pooled}}=16$. Compute Cohen d , interpret, compute Hedges g , 95% CI.

2. Treatment 15/35 events vs control 25/25. Compute OR, RR, ARR, NNT. Clinical benefit?